Royal Devon University Healthcare NHS Foundation Trust

A guideline for the use of Lung Ultrasound by Physiotherapists – Northern Services	
Post holder responsible for Procedural Document	Advanced Respiratory Physiotherapy Practitioner
Author of guideline	Ruth Wood
Division/ Department responsible for Procedural Document	Physiotherapy/ Critical Care
Contact details	Bleep 089 ruthwood4@nhs.net
Date of original policy / strategy/ standard operating procedure/ guideline	015/12/2022
Impact Assessment performed	No
Approving body and date approved	Critical Care Development Group & *Therapy Leads 11.04.2023*
Review date (and frequency of further reviews)	*30/01/2026*
Expiry date	*30/06/2026.*
Date document becomes live	*03/08/2023*

Please *specify* standard/criterion numbers and tick ✓ other boxes as appropriate

Monitoring Information		**Strategic Directions – Key Milestones**	
Patient Experience		Maintain Operational Service Delivery	
Assurance Framework		Integrated Community Pathways	
Monitor/Finance/Performance		Develop Acute Services	✓
CQC Fundamental Standards Regulations No:		Delivery of Care Closer to Home	
		Infection Control	
Other *(please specify)*:			
Note: This document has been assessed for any equality, diversity or human rights implications			

Controlled document
This document has been created following the Royal Devon University Healthcare NHS Foundation Trust. It should not be altered in any way without the express permission of the author or their representative.

Full History		Status: Draft	
Version	Date	Author (Title not name)	Reason
1.0	15/12/22	Advanced respiratory physiotherapy practitioner. FUSIC lung USS mentor and accredited.	*New document, to give guidance for Physiotherapists undertaking Lung Ultrasound*

Associated Trust Policies/ Procedural documents:	*Infection control*
Key Words:	• Lung Ultrasound • Lung USS • LUS • FUSIC • Intensive care unit • Physiotherapist • Physiotherapy • Critical care
In consultation and written with and date: • Gorki Sacher, Consultant Intensivist and ICS Mentor 1/12/22 • Nicholas Love, Consultant Intensivist 1/12/22 **In consultation with and date:** • Gareth Moncaster, AMD for Surgery and Trauma, Critical Care Lead Consultant 02/02/23 • Rebecca Fox, Therapy Research & Innovation Lead 12/01/2023 • ICU clinical governance group 24/2/23 • Helen Cooke Therapies Lead 4/4/23 • Therapy Leads Group consultation till 14/4/23 • Georgina Hands, Respiratory Medicine Consultant consultation till 14/4/23 • Dermot Dalton and Tim Mason, Paediatric Consultants 30/3/23 • Radiology Governance June 2023	
Contact for Review:	Ruth Wood

CONTENTS

1. INTRODUCTION 4
2. PURPOSE 4
3. DEFINITIONS/ ACRONYMS 5
4. DUTIES AND RESPONSIBILITIES OF STAFF 5
5. SCOPE 5
6. USE BY PHYSIOTHERAPISTS 5
7. USE OF FINDINGS 7
8. TRAINING PROCEDURE 7
9. SCANNING PROCEDURE 9
10. EQUIPMENT 9
11. MONITORING AND QUALITY ASSURANCE 9
12. APPENDIX 10
APPENDIX 5: COMMUNICATION PLAN 13
APPENDIX 6: EQUALITY IMPACT ASSESSMENT TOOL 14
13. USEFUL DOCUMENTS 16
14. ARCHIVING ARRANGEMENTS 16
15. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE GUIDELINE 16
16. REFERENCES 17

1. INTRODUCTION

LUS is a well-used and recognised addition to the assessment of the lungs in critically unwell patients. Evidence shows a high diagnostic rate for respiratory failure can be achieved by LUS leading to improved identification and subsequent treatment (Lichtenstein, D., and Mezière, G, 2008). LUS offers significant advantages over both auscultation and CXR as a real-time assessment and monitoring tool (leech et al 2014) and significantly exceeds the sensitivity and specificity profile of CXR (Tierney et al., 2020). A clinical, 9-point LUS protocol strongly agreed with specific CT findings among a diverse population of mechanically ventilated patients with acute respiratory failure (Tierney et al., 2020).

With the availability of portable ultrasound machines, the use of LUS has become more accessible outside of a radiology department and by non-radiologists, with an increasing number of physiotherapists using the technique to facilitate patient management and complement 'usual' physiotherapy assessment techniques. There are currently 91 accredited FUSIC lung ultrasound physiotherapists, 4 in the South West of England (January 2023).

A recent meta-analysis of lung ultrasound in patients with acute onset dyspnoea showed: reduced time to diagnosis, reduced time to treatment, reduced intensive care length of stay by a median of 1.27 days and an improved rate of appropriate treatment (Szabo, G., et al 2022).

2. PURPOSE

This guideline is to provide a framework for the training and use of lung ultrasound (LUS) for physiotherapists. It covers the process to accreditation in lung ultrasound, the uses for respiratory physiotherapists while covering some of the governance issues and practicalities.

3. DEFINITIONS/ ACRONYMS

3.1. LUS: Lung ultrasound

3.2. ICS: Intensive Care Society

3.3. FUSIC: Focussed ultrasound intensive care

3.4. PACS: Picture archiving and communication system

3.5. HCPC: Healthcare Professions Council

3.6. CSP: Chartered Society of Physiotherapy

3.7. I.S: Incentive Spirometry

3.8. MI/E: Manual Insufflation: Exsufflation

3.9. IPPB: Intermittent positive pressure ventilation

3.10. ICU: Intensive Care Unit

3.11. CXR: Chest radiograph

3.12. CT: Computerised Tomography

4. DUTIES AND RESPONSIBILITIES OF STAFF

4.1. Update, review and maintenance of this guideline is the responsibility of the Advanced Respiratory Physiotherapy Practitioner with Lung Ultrasound Accreditation.

4.2. This guideline has been approved by the Critical Care Development Group and Therapy Leads Group.

5. SCOPE

5.1. This guideline applies to physiotherapists working for Royal Devon University Healthcare NHS Foundation Trust, Northern services. It applies to the use of Lung Ultrasound (LUS) to produce images contributing to a respiratory assessment to inform physiotherapy and medical treatment plans.

6. USE BY PHYSIOTHERAPISTS

Background

The use of LUS is increasing and is currently being utilised by both the medical staff and advanced physiotherapists.

The CSP have supported the use of the lung ultrasound as a diagnostic tool with two documents to support staff and aid governance (link in appendix 1.)
Physiotherapists within Northern services with accreditation can utilise LUS when they think it will benefit their physiotherapy assessment and treatment of patients. This is in line with recommendations from a multi centred cohort study which concluded that thoracic ultrasound has a high impact on clinical decision-making process by physiotherapists working in critical care. Le Neindre, A et al 2021 reported a 40 percent change in clinical reasoning by respiratory physiotherapists.

Xirochaki, N et al (2013) also found that from 253 LUS examinations LUS changed management in 119 and 38 percent of the changes were related to respiratory physiotherapy. LUS has a significant clinical impact on decision making and clinical management not only in ICU, the emergency department but on the general wards (Xirouchaki, N., et al, 2014 and Heldeweg, M et al., (2022).

6.1. Indications for Use of LUS for Physiotherapists

Use of LUS may be indicated as part of respiratory assessment and to evaluate treatment by physiotherapists:

6.1.1 As an Adjunct to Respiratory Assessment:

- Identifying potential causes of failure to wean from mechanical ventilation.
- To see progression of pathology with sub-sequential scanning as part of a long critical care/ hospital stay.
- When evolving pathology is suspected but not clear e.g., consolidation/ pleural effusion.
- Accurate pathology identification as an advanced evaluative tool to help:

    1. Assess referrals to physiotherapy giving a clearer idea of the presenting condition for treatment. The aim being to reduce unnecessary treatments.
    2. With advanced use and training differentiation between collapsed and consolidated lung can be determined ensuring correct treatment which improves safety and time efficiency (Haaksma, M et al., 2022).
    3. Understand the severity of a pathology in greater detail.

6.1.2 To Support Evaluation of Treatment Efficacy:

LUS may also be used as a real time assessment tool during physiotherapy treatment; for example, evaluating interventions such as those utilised to increase lung expansion, remove secretions or reverse collapse. This may include techniques such as MI: E / cough assist, IPPB, I.S, deep breathing, mobility and prone positioning.

Lung ultrasound scoring may be used when evaluating treatment as part a regional measure of lung aeration and as a semi quantitative outcome measure to evaluate potential treatments such as recruitment, position changes, exercise, effusion drainage, antibiotics and diuretics. It can be used to guide treatment through the literature. For example, in awake prone positioning lung ultrasound scoring after a single prone positioning session (approx. 3 hours) predicts whether a patient is a responder compared to a non-responder to prone position.

This helps physiotherapists in terms of aiding if it is worth continuing with a treatment, but it also aids prediction risk of intubation (Ibarra-estrada, M et al, 2022).

7. USE OF FINDINGS

Findings from LUS should be used to guide the ongoing physiotherapy and medical management of the patient e.g., not to treat, discuss with consultants regarding findings, assess the appropriateness and effectiveness of interventions, help guide physiotherapy interventions, serial scan to show progress.

When pathology that requires further medical management is detected the accredited physiotherapist should communicate their findings clearly and concisely to the patient's current consultant, who may or may not need to take further action. For example, further discussion with the ICU consultant, and microbiology consultant regarding antibiotics in the case of new consolidation, consideration for starting diuretics in the case of pulmonary oedema, consideration of drainage of a pleural effusion, escalating concerns over findings of a lung abscess, or more urgently a pneumothorax. The accredited physiotherapist has a duty of care to escalate any imaging concerns (Smith, M et al, 2022).

LUS is used as an advanced technique in conjunction with advanced physiotherapy knowledge, skills and clinical reasoning. There is no requirement for LUS to be completed by all physiotherapists working within the critical care environment, and no expectation for those completing on-call and emergency duty working to be trained.

8. TRAINING PROCEDURE

8.1. Staff

Physiotherapists using LUS within Royal Devon University Healthcare NHS Foundation Trust, Northern services should have attended a recognised training programme accredited by the Intensive Care Society (ICS): FUSIC or similar standard.

Physiotherapists undertaking a LUS competency programme should have a designated tutor who holds a qualification in LUS and is an approved mentor. This has offered the most supportive structure for physiotherapists to gain accreditation.

Once FUSIC accreditation/competency in LUS is achieved, it is the physiotherapists' responsibility to maintain their skills, knowledge and competence in line with their professional body, accreditation organisation and clinical specialism. The CSP and HCPC have released a document each in order to support physiotherapists within the last year to support physiotherapists with LUS accreditation. On accreditation a copy of your certificate should be shown to your line manager within your personal file and named consultant mentor.

8.2. Training Stages

Stages are outlined below:

8.2.1 Theoretical Training and Introductory Practice:

Accreditation will require completion of theoretical training such as ICS FUSIC lung module or a similar standard. It is recommended for physiotherapists to complete a course at least part run by a physiotherapist in order to gain insight into the uses of LUS for physiotherapists or through locally delivered training

ensuring that the theoretical and knowledge-based aspects of the syllabus are covered.

8.2.2 Supervised Practice:

At least the first 10 scans involving patients will be directly supervised by a professional with training in the use of LUS with FUSIC mentor status. Findings can only be documented under the FUSIC mentor's accreditation at this stage. A LUS reporting form must be completed for each scan (Appendix 3).

Between scan 10 and 30 the learner can scan patients without direct supervision with patient consent but not document findings in the notes. These scans are labelled, and the appropriate forms filled out/ presented to the FUSIC mentor to go towards the portfolio of competency. This protects the medical/ physiotherapy team from acting upon findings at a point in which that person isn't accredited. These are training scans and should not be considered diagnostic until images are reviewed by a mentor. Any potential findings will be discussed with the relevant teams by the mentor.

It is important to ensure that the trainee has demonstrated competence in safety acquiring and storing images.

8.2.3 Completion of the Logbook:

30 scans are completed in the logbook.

An appropriate range of pathologies should be included in order to be able to fill in the competency sheet (see appendix 2).

8.2.4 Assessment of Competence:

The mentor or supervisor needs to confirm competence - e.g., the core skills have been demonstrated and confirmed in the training record.

A Triggered assessment must be completed with their supervisor.

On completion, the relevant paperwork is signed and sent to the ICS. A certification is received on acceptance of your accreditation.

8.3. Being an ICS Mentor

This can be applied for after one year post accreditation of FUSIC lung with the correct portfolio of experience, and knowledge. Application via the FUSIC website with the link in appendix 4.

9. SCANNING PROCEDURE

9.1. Scanning Protocol

Scans should be performed as per the FUSIC lung protocol or similar. See useful documents section 13.1.

Where there are pathological findings, additional points should be scanned.

9.2. Patient Consent

Full explanation of the procedure should be discussed with the patient. If the patient is sedated and ventilated, then an assessment would be a substituted consent and assessed in the patient's best interests.

9.3. Documentation and Image Management

If the ultrasound machine allows upload to PACS, labelling should be placed in situ in a standardised manner to reflect the side and chest wall zone being examined. This may be a selection of videos or pictures. Images will only be uploaded to PACS by accredited physiotherapists.

Some ultrasound machines do not allow upload to PACS. When using these machines images can be saved to the machine in the same standardised manner. Findings under both situations are documented in notes under the 'results' section by accredited physiotherapists. Saving the images allows for review by the consultant or physiotherapy team as they wish/ need.

Images can be deleted off the ultrasound machine once the patient has left either critical care or the acute hospital site.

10. EQUIPMENT

10.1. Equipment

Should be fit for purpose. A visual inspection of equipment should be made prior to use and any concerns reported to the department owning the ultrasound machine. An individual must only use an ultrasound machine which they have received training on.

10.2. Cleaning

The ultrasound machine must be cleaned before and after use. The equipment should be wiped clean using clinell universal wipes. Excess ultrasound gel must also be removed from the patient using standard local policy for patient hygiene. In the case of infectious diseases, the machine should be cleaned with tristel.

11. MONITORING AND QUALITY ASSURANCE

The use of LUS by physiotherapists will be continuously monitored through regular discussion with consultant colleagues on the ward rounds, records of its use (including anonymised images), patient case studies and reflective practice.

The physiotherapist is responsible for maintaining their knowledge and competence in LUS by undertaking regular and relevant CPD. In order to maintain practical skills, it is important that regular LUS examinations are undertaken which involve an appropriate range of pathologies.

12. APPENDIX

Appendix 1: CSP documents: https://www.csp.org.uk/system/files/publication_files/PD138_POCUSContext_e1_Apr22_0.pdf

Appendix 2: Competency Assessment form (see below).

Appendix 3: Lung Ultrasound reporting form (see below).

Appendix 4: ICS mentor/ supervisor info: https://ics.ac.uk/learning/fusic/mentors.html

Appendix 5: Communication plan (see below)

Appendix 6: Quality Impact assessment tool (see below)

APPENDIX 2 - COMPETENCY ASSESSMENT LUNG ULTRASOUND

	Name	GMC no	Job Title
Trainee			
Mentor			
Supervisor			

Competence	Mentor signature
Checking patient's details/entry into machine as appropriate	
Confirmation of indication and check any supportive imaging	
Positioning of patient and machine ergonomically	
Selection of appropriate probe and optimisation of machine settings at each examination point	
Performs scan with probe placed at upper. lower and postero-lateral points bilaterally with probe orientated longitudinally (cephalad-caudad)	
Identification of subcutaneous tissue, ribs and pleura	
Identification of pleural sliding in 2D and M mode	
Recognition of signs of pneumothorax (2D and M mode) including lung point sign	
Demonstration of lung pulse	
Identification of A lines and B lines	
Identification of normal diaphragm, liver and spleen	
Identification of consolidation / atelectasis	
Identification of interstitial syndrome	
Identification of pleural effusion	
Differentiation of pleural thickening and pleural effusion (including sinusoid sign)	
Differentiation of pleural effusion from sub-diaphragmatic fluid	
Description of appearances of transudate and exudate	
Demonstration of how volume of effusion may be estimated	
Use of ultrasound to identify appropriate site for drainage of effusion	
Demonstration of ultrasound guided thoracocentesis (doctors only)	
Performance of technique safely and effectively	
Attention to sterility with respect to procedure, patient and machine	
Adequate documentation and storage of images and scans as appropriate	
Informing patient and reporting findings where appropriate	
Identification of whether a further scan or alternative imaging is indicated	
Cleaning of equipment and storage to minimise damage	

APPENDIX 3 - LUNG ULTRASOUND REPORTING FORM

Patient details		Operator	
Indication			
Image quality	Good	Acceptable	Poor

Findings									
Probe position	Lung sliding		B lines (>2)		Consolidation		Pleural effusion		Other *(enter text)*
Right upper anterior	yes	no	yes	no	yes	no	yes	no	
Right lower anterior	yes	no	yes	no	yes	no	yes	no	
Right Posterolateral	yes	no	yes	no	yes	no	yes	no	
Left upper anterior	yes	no	yes	no	yes	no	yes	no	
Left lower anterior	yes	no	yes	no	yes	no	yes	no	
Left posterolateral	yes	no	yes	no	yes	no	yes	no	
Other comments (eg size of effusion)									
Conclusion, clinical significance, suggested actions, referral required?									

Signed (trainee)	Signed (mentor)	Mentor comments

APPENDIX 5: COMMUNICATION PLAN

NHS
Royal Devon
University Healthcare
NHS Foundation Trust

The following action plan will be enacted once the document has gone live.

Staff groups that need to have knowledge of the guideline/SOP	*Therapy staff* *Senior staff in ICU*
The key changes if a revised document	*N/a*
The key objectives	*This guideline is to provide a framework for the training and use of lung ultrasound (LUS) for physiotherapists.*
How new staff will be made aware of the procedure/guideline and manager action	*Therapy and critical care staff meetings.* *Email correspondence*
Specific Issues to be raised with staff	*As per guideline.* *Particular attention to information on storing and saving of images and documentation of scans.*
Training available to staff	*As per guideline.* *Courses available through the ICS / British thoracic society or similar standard.*
Any other requirements	*N/a*
Issues following Equality Impact Assessment (if any)	*No negative*
Location of hard / electronic copy of the document etc.	*G drive:OT/SOPS*

APPENDIX 6: EQUALITY IMPACT ASSESSMENT TOOL

Name of document	Guideline for use of the Lung Ultrasound by Physiotherapist
Division/Directorate and service area	Physiotherapy/ Critical Care
Name, job title and contact details of person completing the assessment	Ruth Wood Bleep 089 ruthwood4@nhs.net
Date completed:	01/12/22

The purpose of this tool is to:

- **identify** the equality issues related to a policy, procedure or strategy
- **summarise the work done** during the development of the document to reduce negative impacts or to maximise benefit
- **highlight unresolved issues** with the policy/procedure/strategy which cannot be removed but which will be monitored, and set out how this will be done.

1. What is the main purpose of this document?

This guideline is to provide a framework for the training and use of lung ultrasound (LUS) for physiotherapists.

2. Who does it mainly affect? *(Please insert an "x" as appropriate:)*

Carers ☐ Staff ☒ Patients ☒ Other (please specify)

3. Who might the policy have a 'differential' effect on, considering the "protected characteristics" below? (By *differential* we mean, for example that a policy may have a noticeably more positive or negative impact on a particular group e.g. it may be more beneficial for women than for men)

Please insert an "x" in the appropriate box *(x)*

Protected characteristic	Relevant	Not relevant
Age	☐	☒
Disability	☐	☒
Sex - *including: Transgender, and Pregnancy / Maternity*	☐	☒

Race	☐	☒
Religion / belief	☐	☒
Sexual orientation – *including: Marriage / Civil Partnership*	☐	☒

4. Apart from those with protected characteristics, which other groups in society might this document be particularly relevant to… (e.g. those affected by homelessness, bariatric patients, end of life patients, those with carers etc.)?

Same for all. Likely to be most used with those who are critically ill.

5. Do you think the document meets our human rights obligations? ☒

Feel free to expand on any human rights considerations in question 6 below.

A quick guide to human rights:
N/a

6. Looking back at questions 3, 4 and 5, can you summarise what has been done during the production of this document and your consultation process to support our equality / human rights / inclusion commitments?

Treating all patients with dignity and respect.

7. If you have noted any 'missed opportunities', or perhaps noted that there remains some concern about a potentially negative impact please note this below and how this will be monitored/addressed.

"Protected characteristic":	N/a
Issue:	N/a
How is this going to be monitored/ addressed in the future:	N/a

Group that will be responsible for ensuring this carried out:	Ruth Wood

13. USEFUL DOCUMENTS

FUSIC training document:
https://www.ics.ac.uk/ICS/ICS/Pdfs/FUSIC_Training_Document_2019.aspx (sign in required to access documents).

FUSIC Lung accreditation pack:
https://www.ics.ac.uk/ICS/FUSIC/FUSIC_Accreditation/ICS/FUSIC/FUSIC_Accreditation.aspx?hkey=03ef026c-cd0c-4bba-9555-8c85bb220b01 (sign in required to access documents).

14. ARCHIVING ARRANGEMENTS

The original of this guideline, will remain with the authors: Ruth Wood, Gorki Sacher & Nicholas Love and stored on the G drive in OT/SOPS (Therapy). An electronic copy will be maintained on the Trust intranet, P – Policies – *C – Consent, for example*. Archived electronic copies will be stored on the Trust's "archived policies" shared drive, and will be held indefinitely. A paper copy (where one exists) will be retained for 25 years.

15. PROCESS FOR MONITORING COMPLIANCE WITH AND EFFECTIVENESS OF THE GUIDELINE

To evidence compliance with this policy, the following elements will be monitored:

What areas need to be monitored?	How will this be evidenced?	Where will this be reported and by whom?
Please list the standards that should be monitored i.e. what are the must do's in this document?	*How will you know that the 'must do's' are being done*	*Which groups will monitor the compliance/effectiveness of this document?*
Ensure only accredited physiotherapists upload to PACS	Audit, observed practice, joint scans, feedback	As per the authors
Ensure training scans are only documented in the notes under mentor's accreditation	Observation of notes	As per the authors

16. REFERENCES

Haaksma, M., Smit, J., Heldeweg (2022) Extended lung ultrasound to differentiate between pneumonia and atelectasis in critically ill patients: a diagnostic accuracy Study. Critical care medicine 50 (5): 750-759

Heldeweg, M., Vermue, L., and Brouwer, M (2022) The impact of lung ultrasound on clinical-decision making across departments: a systematic review. The Ultrasound Journal 14 (1): 5

Ibarra- Estrada, M., Gamero-Rodriguez, M., and Acilu, M, (2022) Lung ultrasound response to awake prone positioning predicts the need for intubation in patients with covid-19 induced acute hypoxemic respiratory failure: an observational study. Journal of critical care 26 (2022): 189

Le Neindre, A., Hansell, L., Wormser, J (2021) Thoracic ultrasound influences physiotherapists clinical decision making in respiratory management of critical care patients:a multicentre cohort study. Journal of critical care 31 (2016): 101-109

Leech, M., Bissett, B., and Ntoumenopoulos, G (2014) Lung Ultrasound for Critical Care Physiotherapists: A Narrative Review. Physiotherapy respiratory international 20 (2): 69–76

Lichtenstein, D., and Mezière, G, (2008) Relevance of Lung Ultrasound in the Diagnosis of Acute Respiratory Failure*: The BLUE Protocol Critical Care Medicine 134 (1): 117-125

Smith, M., Hayward, S., and Innes, S (2022) A proposed framework for point of care lung ultrasound by respiratory physiotherapists: scope of practice, education and governance. The Ultrasound Journal 14 (1): 24

Tierney, D., Huelster, J., and Overgaard, J (2020) Comparative performance of pulmonary ultrasound, chest radiograph, and CT among patients with acute respiratory failure. Critical Care Medicine 48 (2):151-157

Xirouchaki, N., Kondili, E., and Prinianakis, G (2014) Impact of lung ultrasound on clinical decision making in critically ill patients. Intensive care medicine 40 (1): 57-65